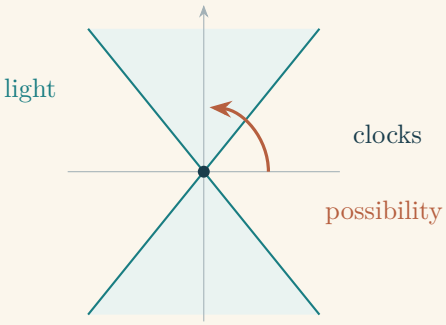
A GUIDE FOR THE CURIOUS

The Universe Has a Complex Side

A Curious Reader's Guide to Complex Relativity



No calculus required.
One imaginary unit, carefully supervised.

Gauss

Before you board

You do not need a physics degree for this trip. You need a willingness to imagine a clock, follow a beam of light, and occasionally let a familiar word mean something unfamiliar.

The destination is a research framework called *Complex Relativity*, or CR. It asks whether a carefully constrained complex geometry can extend our description of spacetime while respecting the real clocks and signals with which we explore it.

We will meet three kinds of ideas along the way. **Established physics** supplies the starting points: relativity, fields, and quantum experiments. **CR's proposals** add particular assumptions about complex geometry and matter. **Open questions** are the places where a proposed connection still needs a derivation or an experimental test. The chapter labels will keep track of which ground we are standing on.

The CR-specific account follows the accompanying research synthesis and its teaching edition.[1, 2] It is a guided tour of that program, not an announcement that its proposed new physics has been confirmed.

There are very few equations. The one that captures the central geometric idea is waiting politely near the back. It will not jump out from behind a diagram.

Gauss

November 2026

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ESTABLISHED PHYSICS

1 The clock on the train

You are on a smoothly moving train, trying to persuade a biscuit to survive a cup of tea. Outside, the scenery races past. Inside, the tea pours normally. A dropped spoon falls by your feet. Provided the train is moving steadily in a straight line, ordinary experiments inside the carriage do not reveal its speed relative to the station.

The station is not the universe's official resting place. Neither is the train. This is the everyday doorway into the principle of relativity: the basic laws of physics take the same form in suitably moving laboratories.

Einstein added a demanding fact about light. In vacuum, inertial observers measure the same light speed even when they move relative to one another. Light does not behave like a tennis ball thrown from the train. The universe has declined the obvious velocity-addition scheme.

Here is a clock built to expose the consequences. Put two mirrors opposite each other and let a light pulse bounce between them. Count each round trip as one tick. Beside the clock, the pulse goes straight across and back. In a frame where the clock moves sideways, it follows a longer zigzag between the moving mirrors. With the same light speed, that longer trip takes more time. The moving clock ticks more slowly in that frame.[3]

This is a comparison using a reference frame and a timing procedure, not simply the delayed image arriving at somebody's eyes. Each inertial observer can consistently assign slower ticking to the other's moving clock. They disagree about which distant events count as simultaneous. There is no universal station clock hiding backstage.

The clock is a physical object

In relativity, an *observer* need not be a person in a white coat. It can be a spacecraft carrying a clock, a detector, and a way to exchange signals. Consciousness is not part of the job description. Calibration is.

A clock follows a path through spacetime. The time it records along that path is called *proper time*. Different paths between meetings can accumulate different proper times. These are differences between physical histories, not failures of punctuality.

CR begins here, with real readings made by real apparatus. Whatever extra mathematics it introduces must make sense of those readings. A theory cannot win by explaining why your watch was imaginary all along.

> **Keep this picture** A clock measures the time along its own journey. Light signals help observers compare journeys. Neither object is an optional decoration added after the theory is finished.

ESTABLISHED PHYSICS

2 Light draws the neighbourhood

Imagine a camera flash at a particular place and moment. Physicists call that combination an *event*. An event is not necessarily exciting. A raindrop hitting a pavement qualifies without selling a single ticket.

After the flash, light spreads outward. At a later moment, it has reached a sphere of places. Stack those expanding spheres in order of time, and you obtain a *light cone*. A flat drawing usually shows only one direction of space, so the cone looks like two diagonal lines.

The cone is a map of possible causal connections. Its future interior contains events that a slower-than-light influence could reach from the flash. Its boundary is reached by light in vacuum. Outside it, an event cannot be reached from the flash by an influence confined to that light-speed limit. A past cone reverses the story: it marks events that could have influenced the flash.

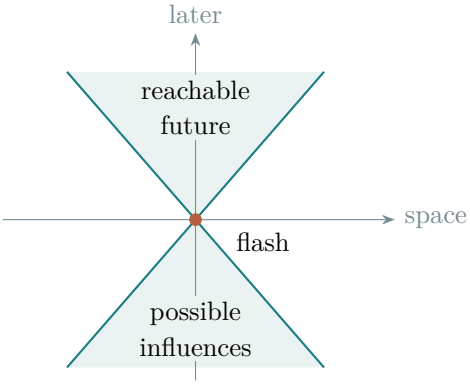


Figure 1: One direction of space, with time running upward. The sloping edges are the light boundary. This is a local spacetime diagram, not a picture of a lamp casting a triangular beam.

Why the technical name is *null*

The spacetime measure that distinguishes lightlike paths assigns them a special value: zero. That is why physicists also say *null cone*. Zero here does not mean that nothing travels or that no distance is covered. Space and time enter this particular measure with different signs, and for light the contributions balance.

A light cone tells us which directions are lightlike. It does not tell us how many seconds a clock should record along a nearby timelike path. Think of a map whose road layout is known but whose scale bar has fallen off. A clock helps restore the missing scale.

In curved spacetime, each event has its own local cone. How those cones fit together determines the wider causal story. There is no requirement that the entire universe fit on one straight-sided diagram like the one above.

CR treats this causal structure as a foundation. Its complex geometry must share the real light directions. That apparently modest demand will soon do a surprising amount of work.

THE MATHEMATICAL TOOLKIT

3 Imaginary numbers have day jobs

The name *imaginary number* is an unfortunate piece of public relations. It suggests a number invented by someone whose homework was due in ten minutes.

The basic imaginary unit, written i, obeys a perfectly precise rule: multiplying it by itself gives minus one. A useful picture is a quarter-turn. On a number plane, multiplication by i turns an arrow through a right angle. Do it twice and the arrow points backward. The rule has become geometry.

A complex number combines an ordinary-number part with an imaginary-number part. Together they behave like a two-coordinate system with a useful rule for multiplication. Physicists use that rule to keep track of oscillations and relative phases.

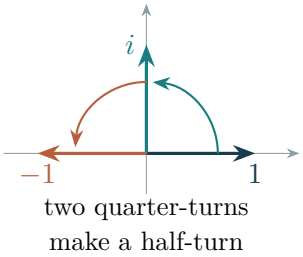


Figure 2: A way to understand $i^2 = -1$. The extra axis is a mathematical coordinate, not evidence for a hidden corridor in your house.

For example, two oscillations can reinforce each other when they are in step and partly cancel when they are out of step. Complex numbers package the strength and timing of an oscillation neatly. Using them in a calculation does not make the sound or voltage coming out of the experiment unreal.

What CR makes complex

In CR, the complex structure belongs to a proposed description of spacetime geometry. The physical events we observe still lie on a four-dimensional real spacetime: three directions of space and one of time. The larger complex mathematical setting is not a claim that we can wander through four additional everyday directions.

Think of coordinates as ways to organize information before treating them as places you could visit. A weather report may specify temperature, humidity, pressure, and wind speed. It has several coordinates describing the weather, but you cannot turn left into humidity.

There is another distinction worth carrying with us. Complex numbers appear in quantum mechanics too. Their appearance in two theories does not make the theories identical. Having the same alphabet does not make a shopping list a Shakespeare play.

The interesting question is what physical rules accompany the complex mathematics. That is where CR's two postulates enter.

CR'S STARTING POINTS

4 Two promises the theory makes

A postulate is a rule a theory agrees to start from. It should be clear enough that we can work out what follows from it and recognize when a model fails to obey it. A useful postulate is a commitment, not a decorative sentence at the front of a paper.

CR has two mathematical and physical foundations: *complex spacetime* and the *real causal light cone*. Its two physical postulates, P1 and P2, connect those foundations to observers and particles.

P1: respect the clocks and the light directions

Physical measurements belong to observers carrying clocks on real worldlines. Local freely moving laboratories obey the same basic kinematic laws. Observers can accelerate, but their motion can still be compared with a local freely moving frame at each event.

The extra CR requirement is that every direction marked as lightlike by real measurements must also be null for the *full complex* geometry. Checking only the real part would be too easy: the imaginary part could then distort the same directions without anyone asking it a difficult question.

This compatibility requirement is an added premise of CR. Ordinary relativity alone does not force an independently introduced complex geometry to obey it.

P2: keep physical momentum and mass real

The second postulate starts with real, measurable particle momentum and a nonnegative real mass. Momentum tells us about motion and energy; its allowed values are related to the particle's mass. Physicists call this relation a *mass shell*. No shellfish are involved.

CR extends that momentum into the complex mathematical setting in a natural way: when it is paired with a real physical displacement, it returns the original real result. The complex mass-shell expression must then agree with the same physical relation, rather than add an unwanted second restriction.

The compatible construction makes the real and imaginary parts of the complex equation two versions of the same condition. Think of one bank balance displayed in two currencies, with the conversion specified. Two displays do not create two independent balances. In the mathematical construction, the corresponding coefficient is fixed so that the two conditions vanish together.

The physical motion is generated by a real dynamical rule. Mass does not become imaginary, and a clock is not asked to read a complex number on its dial.

These promises are strong enough to constrain the theory. They do not yet specify every interaction, every particle, or the rules of quantum measurement. For those, we must build and test further pieces.

A CONSEQUENCE OF P1

5 Ten knobs become one

A metric is a rule for turning coordinate information into spacetime intervals. Coordinates tell you where to put the labels on a map. The metric tells you what those labels mean for measurements.

In four-dimensional spacetime, a general symmetric metric has ten independent entries at each event. Four describe the individual coordinate directions; six describe how pairs of directions mix. CR initially allows an imaginary part with its own ten entries. Imagine ten adjustable knobs, each able to alter some aspect of the proposed complex geometry.

Now impose P1's requirement: *every* real light direction must remain null for the full complex metric.

Point the test light direction one way, then the opposite way. Their comparison removes certain directional mixtures. Test the coordinate axes and diagonals, and further distortions disappear. The full proof is short enough for the teaching edition, but the result is easy to state: the imaginary part must have exactly the same tensor pattern as the real metric, multiplied by a single real number.

The ten imaginary knobs have become one.

One setting at each event

That remaining number is called χ, pronounced roughly *kai*. It may have different values at different events. The requirement fixes its geometric role, not its numerical value everywhere.

Why can this freedom survive? A null interval is zero, and multiplying zero by a common factor still gives zero. The cone tests distortions of light directions very efficiently, but cannot detect a factor multiplying the entire interval.

The map analogy helps here: enlarging a road map preserves which roads meet, although distances measured on the paper change. The complex case is more precise than an ordinary zoom. The physical real metric and its clock calibration are retained, while the imaginary part is forced to follow the same pattern. We are not simply enlarging everybody's clocks.

A theorem, with its premises visible

This reduction is a mathematical consequence of the stated compatibility condition. It does not establish that nature has chosen the condition. That is the difference between proving what a proposed rule implies and proving that the rule describes the universe.

Still, it is a useful result. A large collection of possible imaginary distortions has been reduced to a single field. The theory has become easier to analyze, and its remaining freedom is specific enough to be given dynamics.

The next question is what that lone surviving knob does when we allow it to change.

INSIDE A CHOSEN CR MODEL

6 When a number learns to move

A field is something that assigns a value to each place and time. The temperature in a room is an everyday example. You do not have to picture a field as a fog or a fabric. It is a structured way of describing local physical quantities.

The surviving CR variable χ can be treated as a field. But saying that a field exists is not yet an explanation of its behavior. A weather map does not come with tomorrow's forecast included. We need a rule for change.

Physicists often supply that rule with an *action*. An action assigns a number to a possible history of a system. The physical equations follow by asking which histories make that number stationary under small allowed changes.

The universe need not audition every possible future and compare spreadsheets. The action is a compact way for us to write the equations. Also, stationary does not always mean smallest. Physics has enough subtleties without hiding a minimization contest in every sentence.

A familiar result from an unfamiliar route

One CR model takes a complex version of the gravitational curvature action, restricts it to the geometries allowed by P1, and keeps the physical action real. This choice reduces exactly to Einstein gravity coupled to an ordinary, positive-energy scalar field.

A *scalar* is a field with one value at each event, without an attached spatial direction. The useful canonically normalized version is called ϕ, pronounced *fee*. It is a different coordinate for the same surviving geometric freedom, not a second new field.

The change from χ to ϕ is like replacing an awkwardly marked measuring scale with one that makes the motion law simple. The underlying configurations have not multiplied.

In this model the field can vary, carry energy, and contribute to gravity. If it is constant and the other sources permit it, the familiar vacuum solutions of general relativity reappear. That is a useful correspondence with established physics.

A recipe has been chosen

The two postulates did not uniquely demand this action. Other compatible interactions can be written, and different choices can lead to different predictions.

This is where theory-building becomes concrete. State the recipe, work out its consequences, and check whether the result is stable and physically useful.

The scalar's positive kinetic energy is encouraging. It is not an automatic solution to dark matter or dark energy. A new ingredient does not become dessert simply because we have placed it in the cosmic kitchen.

To find out how it interacts, we need the response sectors.

INSIDE A CHOSEN CR MODEL

7 Meet c: the other c

Physics has reused the letter c. There should probably have been a committee.

The famous c is the speed of light. The c in CR's *response sectors* is a different quantity: a dimensionless parameter controlling how a pair of fields responds to the shared geometry. Changing this parameter does not turn a light-speed dial.

Each response pair has two real components. Their motion can be recombined into two independent directions at a fixed background. The parameter c controls the energy weights of those directions.

Setting	What happens to the two kinetic weights
$c > 1$	Both are positive: the normal sectors.
$c = 1$	One is positive and one vanishes: the critical sector.
$-1 < c < 1$	One is positive and one negative.
$c = -1$	One vanishes and one is negative.
$c < -1$	Both are negative with the chosen overall action sign.

Positive weights allow the usual increase in motion energy as a mode is excited. A negative kinetic direction is troublesome when coupled to positive-energy matter: the model can allow energy to fall without a lower bound. This is a warning about the theory, not a proposal for a household generator.

The special point $c = 1$ needs different mathematics. One kinetic coefficient disappears, so we must determine which motions are still independent and which are constrained. A missing coefficient is not automatically a new freedom. Disconnecting a control from a dashboard does not prove the vehicle can now fly.

What is special about 2, 3, and 4?

The research models emphasize the positive settings $c = 2, 3, 4$. At a common reference background, their pairs of weights are respectively $3, 1$; $4, 2$; and $5, 3$. The simple pattern is useful for building related models. The mathematics of positivity also permits noninteger values.

The labels alone do not identify particles. Setting $c = 1$ does not manufacture a photon. Setting $c = 3$ does not, by itself, produce the strong force. Additional fields, symmetries, interactions, and physical tests are needed. Likewise, $c = 0$ contains a negative response direction; it is not a shorthand for pure gravity.

What the positive sectors do share is the geometric scalar. That common connection creates relationships between their interactions. To understand those relationships, we need one more kind of geometry.

GEOMETRY OF A MODEL

8 A landscape made of settings

Imagine a sound desk with several sliders. One controls a voice, another a violin, another the bass. The collection of all slider settings forms a *space of configurations*. Moving through that space means changing the sound, not walking across the room.

Fields have a similar space of possible values. In the selected CR model, three response pairs contribute six settings, and their shared scalar contributes one more. Together they form a seven-dimensional *field space*, also called a target space.

The physical world in this model still has three spatial dimensions and time. Seven settings at each event do not make seven directions along the pavement.

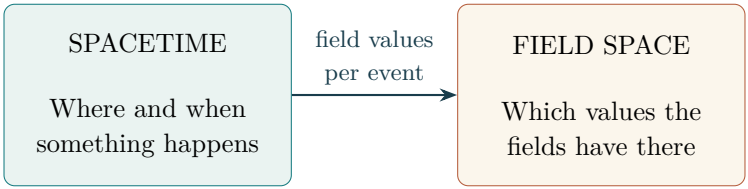


Figure 3: Two different jobs for geometry. A point in field space describes a configuration; it is not an extra physical address.

The sliders are coupled

Suppose adjusting one control changes how the others respond. You may be able to recalibrate the controls around one setting so that everything looks simple. Across the whole range, however, the couplings may resist every attempt to remove them by relabeling the sliders.

Curvature is the mathematical test for this obstruction. The normal CR response spaces have nonzero curvature. Their interactions cannot all be turned into those of free, independent fields merely by choosing clever names for the variables.

The single geometric scalar can be given a simple kinetic coordinate, as we saw earlier. Once the response fields are included, the full configuration space is curved. Simplifying one slider has not simplified the entire desk.

Several sectors sharing the same scalar also differ from several unrelated sectors, each with its own scalar. It is the difference between instruments on one vibrating platform and instruments on separate platforms. Motion can be communicated through the shared structure. The equations determine the actual interaction; the platform is only a picture to help us remember its origin.

> **A minus sign with a different job** The normal target spaces have negative scalar curvature. That describes their geometry. Their kinetic energy can still be positive. Negative curvature and negative energy are different statements about different quantities.

Why counting components is not enough

The broader CR program also builds internal spaces carrying the symmetries used in particle physics. A specified three-plus-two internal splitting can organize the familiar color, weak, and hypercharge structure.

That splitting is extra mathematical input. Its number of components is not a substitute for proving the right particle properties. Three knobs are not necessarily three quark colors, any more than three strings guarantee that an instrument is a violin.

A POSSIBLE APPLICATION

9 Dark matter needs more than a name

Astronomers infer more gravitating matter than they can account for by adding up the ordinary material they observe. The clues include the motions of galaxies and the bending of light by large concentrations of mass.

That bending, called *gravitational lensing*, helps map where mass is distributed even when it does not shine. The evidence concerns gravity and cosmic structure; it does not arrive with a photograph identifying a new particle. *Dark matter* names the inferred component whose physical nature has to be worked out.[4]

This makes dark matter an inviting target for new theories. It also makes it a crowded target. Naming a field after an astronomical mystery is the beginning of an application, not the end of an investigation.

What the $c = 4$ model offers

One CR model uses the positive $c = 4$ response and gives one combination of its fields a mass. Near the bottom of its quadratic potential, that field can oscillate much faster than the universe expands. On a slowly changing geometric background, its energy, averaged over many oscillations, can dilute with cosmic expansion much like cold matter.

This is a meaningful result: the model has a branch with the right kind of large-scale behavior. It does not determine how much of the field was present initially or prove that it produces the structures we observe.

There is a further complication. The same simple construction also leaves another response combination massless. Setting that field to zero in a background calculation does not remove the possibility of exciting it later.

The decay nobody invited

A symmetry can forbid certain decays. For instance, a rule that divides particles into two classes can protect the lightest particle in one class, provided every interaction respects that rule.

The phrase *lightest particle* does a lot of work. A candidate protected in a small isolated model may lose that protection when more sectors are added. New light particles can supply allowed final states. A shared mediator can even provide an indirect decay route when a simple direct route is closed.

The emergency exit was there all along; the building inspector had only checked one room.

For a dark-matter candidate, we must calculate the full spectrum, possible decays, production, lifetime, and gravitational clustering. Being electrically neutral and mathematically well behaved is a promising start, not a completed application form.

Dark energy is a separate assignment

The free homogeneous geometric scalar in the minimal CR gravity model loses energy with expansion faster than ordinary cold matter. It does not automatically behave like dark energy.

A constant or sufficiently slowly changing potential can provide a different cosmic behavior, but that potential is additional input. Dark matter and dark energy have different jobs. The word *dark* is not a universal employment contract.

THE QUANTUM QUESTION

10 Complex does not automatically mean quantum

Quantum mechanics uses complex amplitudes, and those amplitudes interfere. CR uses complex geometry. It is tempting to put these two facts together and declare the quantum mystery solved.

Tempting, but premature.

Ordinary classical waves also have phases and interference. They can be described with complex numbers. Some of their intensity patterns even have the same mathematical shape as familiar quantum probability formulas.

So a complex description, a wave pattern, or an attractive geometric overlap is not enough to reconstruct all of quantum physics. A tuxedo is not proof that the penguin has become a waiter.

What else is needed?

A physical theory must say what states can actually be prepared, what quantities can actually be measured, and what probabilities those measurements have.

In quantum mechanics, some pairs of observable quantities are represented by mathematical operations that do not commute: applying them in one order differs from applying them in the reverse order. The quantum state must also assign consistent, nonnegative probabilities to every allowed measurement, including combinations that are more demanding than one attractive two-outcome example.

The familiar uncertainty principle belongs to that structure. For suitable pairs, such as position and momentum, the spreads in a prepared quantum state cannot both be made arbitrarily small. This is not merely a complaint about clumsy laboratory equipment.

CR's geometry contains ingredients that make these questions interesting. Its response matrices can be connected with mathematical structures used for simple quantum systems. Its fields can interact with apparatus. Its reductions can produce useful phase-space geometries.

What remains is to show that the physically preparable states and readable quantities have the full required quantum structure. A mathematical resemblance must become an account of experiments.

The scale also matters

Planck's constant sets the scale of quantum action. A classical oscillator can have a well-defined rhythm without selecting that quantum scale. Knowing the tune does not tell us how the tickets are denominated.

One approach is to take a CR action and quantize it using the established rules of quantum theory. Another, more ambitious approach is to explain why those rules and their common action scale arise in the first place. These are different research goals.

The booklet does not ask you to choose a winner in advance. It asks you to recognize the difference between finding a useful mathematical ingredient and completing the physical recipe.

MODELLING OBSERVATION

11 The universe reads its own instruments

A theory of the whole physical world has to put its measuring instruments inside the world. There is no neighbouring universe from which to borrow a perfectly passive ruler.

Thermometers exchange energy. Detectors absorb or redirect signals. Clocks need a reliable process, and records must last long enough to be read. An unattended detector can do all this while everyone is at lunch. Observation is a physical process.

A pointer gets a push

The CR response interactions can couple a system to a pointer. One field moves the pointer, and the pointer pushes back. This models signal transfer and measurement backreaction.

Backreaction alone does not derive quantum uncertainty: classical instruments also disturb their targets. We must specify how the pointer is prepared and what it reads.

A lasting record needs more than a momentary deflection. It might use a latch with two stable positions, like a switch that stays put. The model must explain what triggers it, where energy goes, and how noise affects reliability.

Why calibration changes probabilities

Consider two independent detector channels in a *memoryless* model. Before clicking, each has a fixed chance of clicking in the next short interval, regardless of how long it has waited.

Under this statistical assumption, signal weights of one to three give the first channel a one-quarter share of recorded races when sensitivities are matched.

Double its effective sensitivity. Rates become two to three; its share becomes two fifths, or forty percent. The incoming signal has stayed the same. The apparatus has changed.

A probability formula therefore needs a preparation and detection model. A fuller account must also predict waiting times, missed events, and the effect of one measurement on the next.



Figure 4: A schematic apparatus. Transferring a signal, disturbing a system, and retaining a record are related but distinct tasks.

A clock has to keep being a clock

An oscillating field can serve as an internal clock while its phase—the position within each cycle—advances reliably. If the oscillation disappears, the reading fails. An instrument cannot tell time by confidently pointing at an undefined angle.

We can ask what a pointer reads when this clock reaches a specified phase, including the clock's energy and influence in the model.

The aim is to build a workable laboratory from the same ingredients as its subject.

WHAT COMES NEXT

12 How to make a beautiful idea nervous

A beautiful theory should eventually face a measurement it cannot flatter.

The first test is internal: are its equations consistent? Do allowed disturbances evolve sensibly? Is the energy well behaved? Do physical signals respect the causal structure the theory promised? If an interacting model violates its own postulates, the solution is to repair or reject that model.

Then come distinctive physical predictions. The shared scalar in CR ties some interactions together. Within a specified model, their relative strengths can be calculated. A successful measurement of several linked effects would be more informative than a fit achieved by independently adjusting each one.

But the qualifier *specified model* matters. Additional allowed interactions and quantum corrections can alter simple relationships. A prediction must travel with its assumptions, rather like luggage with a name tag.

Several ways the program could advance

A dark-sector model could calculate a complete history of production, decay, and cosmic clustering. A critical-sector study could settle which disturbances really propagate. A gauge construction could explain why its internal structure is selected instead of assuming the desired pattern at the start.

A quantum reconstruction could derive the actual observable quantities, the allowed states, the common action scale, and the behavior of combined systems. Single-particle analogies are useful; entangled systems and sequences of measurements ask much harder questions.

Conversely, a failed calculation can be valuable. It may show that a simple claim was too strong, that a supposedly stable particle can decay, or that a proposed measurement rule depends on a hidden calibration assumption. Finding the limit of an argument is progress when the limit is stated clearly.

What you can now recognize

CR begins with real clocks and signals, introduces complex geometry, and demands that the two descriptions share the real light directions. That demand reduces many possible imaginary distortions to one scalar freedom. Compatible momentum rules preserve real particle physics at the free level. Additional actions give the scalar and response sectors their dynamics.

The result is a concrete research framework: enough structure to calculate, enough unresolved physics to investigate.

You do not have to choose between enthusiasm and careful judgment. Curiosity supplies the questions; calculations and experiments decide how far the answers travel.

The universe may have a complex side. Whether this particular map describes it is a question the universe gets to answer.

One equation, strictly optional

If you would like to see the central geometric result in its native habitat, here it is:

$$\boxed{\gamma = (1 + i\chi)\, g.}$$

Read it as a sentence:

> The complex metric is the real physical metric multiplied by one common complex factor.

The symbols have modest jobs:

- g is the real metric used for physical spacetime measurements.
- γ, the Greek letter gamma, is the induced complex metric.
- i is the imaginary unit.
- χ is the single real field that controls the imaginary part.

The real part of the factor is one, so the real part of γ is still g. A direction that gives zero interval for g also gives zero for γ. The stronger result runs the other way: under the assumptions used in the CR construction, requiring compatibility for *all* real light directions forces this form.

The equation does not specify how χ changes, which particles exist, or how probabilities arise. Those require the further physical work described in the booklet.

You have now met the equation. It was shorter than its introduction.

A pocket glossary

Action
A mathematical rule assigning a number to a possible history; its stationary variations give equations of motion.

Complex number
A number with real and imaginary parts, with a multiplication rule useful for describing rotations and phases.

Field
A quantity with a value at each event in spacetime.

Field space
The space of possible field values. Its directions describe changes in configuration, not extra walking directions.

Light cone, or null cone
The local lightlike boundary separating possible causal directions from those outside the light-speed limit.

Mass shell
The allowed energy–momentum relation for a particle with a specified mass.

Metric
The rule that turns coordinate information into geometric intervals.

Observer
A physical system able to keep time and compare events through signals and records.

Postulate
An explicit starting assumption whose consequences a theory agrees to follow.

Proper time
The time accumulated by an ideal clock along its own timelike path through spacetime.

Response sector
A selected set of fields and their coupling to the common geometry. Here its parameter c is not the light speed.

Scalar
A quantity specified by a single value at an event, without a spatial direction attached.

Symmetry
A transformation that preserves a specified physical structure or the laws of a model.

Target curvature
Curvature of a field-value space, which can encode interactions that cannot be removed by relabeling fields.

Sources and next stops

This booklet adapts the accompanying CR research into an account for general readers. The train, map, dashboard, and sound-desk comparisons are explanations, not additional physical postulates. Each analogy is useful only for the feature it illustrates.

The detailed mathematical account is in the first two references below. The last two are independent background reading on established relativity and astronomical evidence for dark matter; they do not endorse CR's proposed extensions.

References

[1] *Introduction to Complex Relativity*, lecture notes, 3 November 2026. The accompanying 44-page guide contains derivations, worked examples, exercises, and solutions.

[2] *CR Theory from P1 and P2: Complex Spacetime, the Common Null Cone, and Physical Dynamics*, revised research synthesis, 2 November 2026. The direct technical source for the particular CR framework discussed here.

[3] Markus Pössel, *From light clocks to time dilation*, Einstein Online, Max Planck Institute for Gravitational Physics. Read the illustrated light-clock explanation.

[4] NASA Science, *Dark Matter*. Explore the astronomical evidence and gravitational lensing.

The question to take away: If a new mathematical description respects our clocks and light signals, what further experiments would show that its extra structure is physically real?